

Multi-Discriminator Active Adversarial Network for Multi-Center Brain Disease Diagnosis

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Abstract—Multi-center analysis has attracted increasing attention in brain disease diagnosis, because it provides effective approaches to improve disease diagnostic performance by making use of the information from different centers. However, in practical multi-center applications, data uncertainty is more common than that in single center, which brings challenge to robust modeling of diagnosis. In this article, we proposed a multi-discriminator active adversarial network (MDAAN) to alleviate the uncertainties at the center, feature, and label levels for multi-center brain disease diagnosis. First, we extract the latent invariant representation of the source center and target center to reduce domain shift by adversarial learning strategy. Second, the proposed method adaptively evaluates the contribution of different source centers in fusion by measuring data distribution difference between source and target center. Moreover, only the hard learning samples in target center are identified to label with low sample annotation cost. Finally, we treat the selected samples as the auxiliary domain to alleviate the negative transfer and improve the robustness of the multi-center model. We extensively compare the proposed approach with several state-of-the-art multi-center methods on the five-center schizophrenia dataset, and the results demonstrate that our method is superior to the previous methods in identifying brain disease.

Index Terms—Multi-center analysis, adversarial domain adaptation, active learning, brain disease diagnosis.

I. INTRODUCTION

NEUROIMAGING has been widely applied into brain disease diagnosis, since it could reveal the underlying pathological mechanism of brain diseases in the non-invasive way [1]. With the development of machine learning techniques, many

intelligent brain disease diagnosis methods using neuroimaging have been proposed [2], [3], [4], [5]. However, most of these methods are trained and tested within the single center dataset [6], which restricts the generalization of the diagnostic model. The number of subjects from a single center is often limited, and there are differences in data distribution among centers, which means it is difficult to establish a robust brain disease prediction model with a single center. Therefore, it is highly essential to investigate effective multi-center modeling methods to fuse the data across centers to improve the diagnostic performance.

Multi-center [7] neuroimaging data are often generated and collected in the multi-source heterogeneous distributed environment with a large spatial and temporal scale, resulting in high data uncertainty [8], [9]. In a practical multi-center diagnosis problem, data uncertainty often occurs simultaneously at the center, feature, and label levels. For example, variations in imaging platforms or differences in the distribution of demographic characteristics among subjects lead to uncertainty at the center level. At the feature level, the misaligned semantic features among the centers lead to uncertainty in the fusion analysis process. Due to the high cost of annotation, often only a portion of the samples have supervised information, which leads to the uncertainty of the training results at the label level. These uncertainties present a challenging problem in learning robust multi-center diagnosis models.

Some multi-center studies directly combine the samples from different centers for fusion analysis as shown in Fig. 1(a). For example, Leming et al. [10] utilized convolutional neural network (CNN) with the multi-source functional magnetic resonance imaging (fMRI) connectome data for the autism diagnosis. Heinsfeld et al. [11] applied deep learning algorithms to identify autism spectrum disorder (ASD) patients from a large multi-center brain imaging dataset based on the patient's brain activation patterns. However, these methods ignore the uncertainty among different centers, which leads to generalization degradation in brain disease diagnosis.

In recent years, many researchers applied the multi-source domain adaptation (MSDA) to reduce the uncertainty at feature level, which makes that the features from different centers are aligned or comparable, as shown in Fig. 1(b). For example, Wang et al. [12] proposed a multi-center adaption framework via low-rank representation decomposition (maLRR) for ASD identification using fMRI. Shi et al. [13] introduced a three-way decision theory into unsupervised domain adaptation, and used

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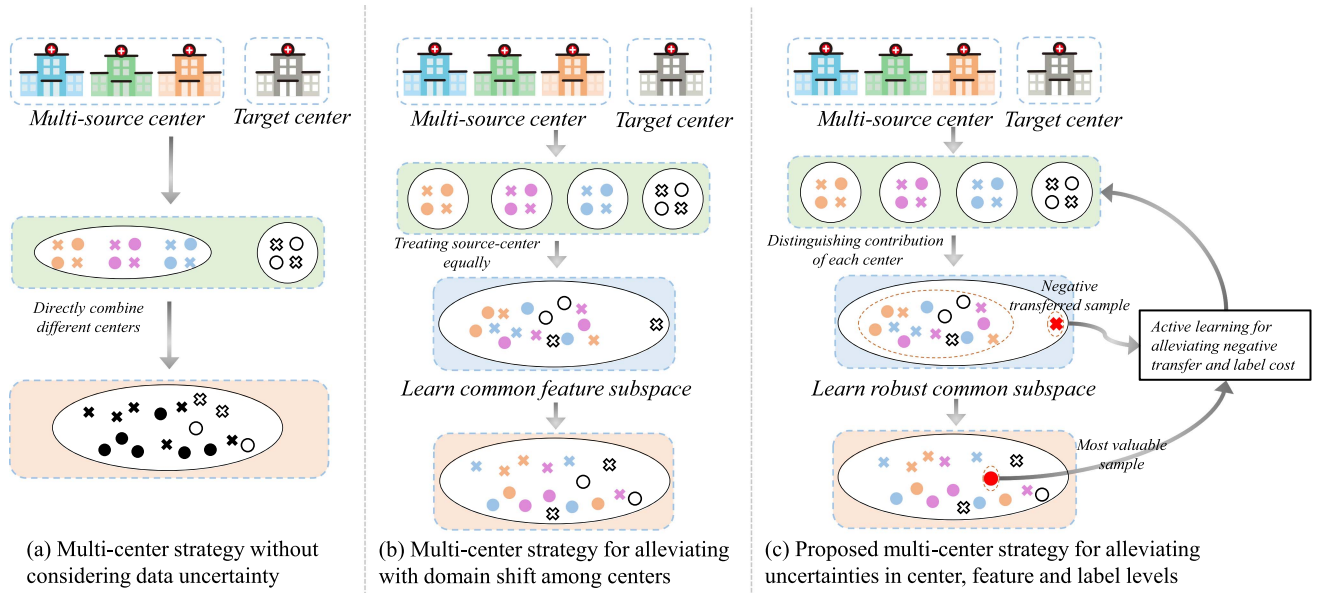


Fig. 1. Three different strategies of multi-center brain disease diagnosis. (a) combines the multi-center data together directly for training, which ignores the data uncertainty. (b) learns a common feature subspace for alleviating uncertainty at feature level. (c) is our proposed multi-center strategy, which can both learn a discriminative subspace by distinguishing the contribution of each center and mitigate the negative transfer through iteratively selecting the most valuable samples.

it to optimize the pseudo label of the target domain site for classifying ASD patients. Most of the existing unsupervised domain adaptation methods try to achieve the aim of transferring knowledge by exploring domain-invariant structures and representations among different centers [14], [15]. However, these approaches ignore the uncertainty from the influence of hard learning samples on model generalization, which often appears in real multi-center applications.

To alleviate above data uncertainties in multi-center modeling, a robust multi-discriminator active adversarial network is proposed in this paper. This framework learns the robust common feature representation among centers, reduces the influence of negative transfer samples, and improves model discrimination with low annotation cost. Compared to the conventional domain adaption strategy, our method can alleviate the uncertainties at multi-center problems at center, feature, and label levels, so it is more appropriate for the practical application of brain disease diagnosis tasks. First, we develop an adversarial learning network to extract the latent invariant representation of the source and target domains. Second, the fused multiple discriminators are utilized to align the features of the multi-center source domains. Then, by introducing active learning in above model, we adaptively identify the hard learning samples to reduce their negative transfer influence on the model. Finally, we employ the selected samples in the target center as auxiliary data to perform semi-supervised learning to make diagnosis decisions.

In brief, the proposed multi-center analysis framework has the following contributions:

- 1) We learn the invariant feature representation among different domains by adversarial domain adaptation, which reduces the heterogeneity between the target domain and source domain.

- 2) The hard learning samples are iteratively selected to label by active learning strategy, and they are treated as the auxiliary domain to improve the robustness with low sample annotation cost.
- 3) The proposed method effectively fuses the information from different centers by adaptively evaluating the contribution of each source center to the target center.
- 4) The experimental results show the proposed method outperforms the state-of-the-art methods on multi-center schizophrenia diagnosis tasks.

The rest of this paper is organized as follows: Section II introduces related work; Section III describes the studied materials and our proposed method; The experimental results and discussion are presented in Section IV; Section V concludes this paper.

II. RELATED WORKS

A. Resting State fMRI Functional Connectome

In recent years, resting state fMRI is the widely used imaging technique for the study of the human brain's functional connectivity. Many researchers have taken advantage of the resting state fMRI functional connectome to enhance human comprehension of brain health and disease [16].

It has been shown the effectiveness of using machine learning techniques with the resting state fMRI functional connectome in the classification of mental disorders, such as Autism [12], schizophrenia [17], and Alzheimer's disease (AD) [18]. For example, Li et al. [19] employed least absolute shrinkage and selection operator (LASSO) to estimate hyper-connectivity for characterizing functional interaction among multiple brain regions for mild cognitive impairment (MCI) classification. In

recent years, a large number of studies have used convolutional neural networks to predict brain diseases in resting state fMRI functional connectome. Ju et al. proposed to use an autoencoder network to distinguish normal aging from mild cognitive impairment, which also reveals discriminative biomarkers for AD detection [20]. Zhu et al. proposed a community detection and convolutional autoencoder network to jointly learn multi-modal brain image features for brain disease diagnosis [21].

B. Unsupervised Domain Adaptation

Single-source unsupervised domain adaptation (UDA) can handle situations, where a model is trained on labeled data from a source domain and unlabeled data from a related but different target domain. It can improve the robustness of the model by making full use of the correlation of samples in the target domain [22]. Researchers have proposed several metrics to measure the differences between domains, such as Maximum Mean Discrepancy, Kullback-Leibler Divergence, and Jensen-Shannon Divergence [23], [24]. In general, they optimize these metrics to reduce the domain shift among centers.

For the past few years, with the booming development of Generative Adversarial Networks (GAN) [25], researchers have proposed to construct an adversarial loss to accommodate the domain shift by introducing domain-classifier [26]. In the optimization strategy of adversarial loss, the domain classifier is trained to discriminate whether input features are from the source or the target, whereas the feature extractor is trained to deceive the domain classifier into matching feature distributions. Long et al. [27] presented conditional adversarial domain adaptation, a principled framework that conditions the adversarial adaptation models on discriminative information conveyed in the classifier predictions. Zhao et al. [28] proposed a multi-source distilling domain adaption (MDDA) method by measuring the similarities of different samples across domains. Li et al. [29] proposed an adversarial training-based sensor-invariant feature extraction method with real industrial applications. Tang et al. [30] proposed a novel adversarial learning method, i.e., Discriminative Adversarial Domain Adaptation (DADA), to overcome the limitation in aligning the joint distributions of feature and category across domains.

While it is feasible to optimize domain alignment and class discriminability simultaneously, finding a balance between them is critical and can significantly affect the performance of the model. Xiao et al. [31] proposed dynamic weighted learning (DWL) to avoid the discriminability vanishing problem caused by excessive alignment learning and domain misalignment problem caused by excessive discriminant learning. However, the adversarial discriminator for feature confusion may also be inaccurate. In some situations, even if the discriminator is fully confused, the sufficient similarity between two distributions cannot be guaranteed. To solve this problem, Gao et al. [32] demonstrated that the distribution discrepancy can be reduced by constraining feature gradients of two domains to have similar distributions.

In practical applications, the situations of target domain using multi-source data for training are becoming more and more

TABLE I
DEMOGRAPHIC CHARACTERISTICS OF PARTICIPANTS

Dataset	Group	Age	Gender(M/F)
NBH	Normal	35.29 ± 7.94	10/14
	Patient	30.78 ± 9.01	6/15
COBRE	Normal	34.82 ± 11.28	46/21
	Patient	36.75 ± 13.68	42/11
Nottingham	Normal	33.38 ± 8.98	95/85
	Patient	33.34 ± 9.05	27/5
Taiwan	Normal	29.87 ± 8.62	25/37
	Patient	31.59 ± 9.60	35/34
Xiangya	Normal	27.17 ± 6.64	35/25
	Patient	23.37 ± 7.83	49/34

common [33]. For example, Zhang et al. [34] proposed a decentralized federated transfer learning method, which achieved promising prediction results in collaborative fault diagnosis. Peng et al. [35] proposed the moment matching-based multi-source domain adaptation method (M3SDA), which aims to transfer knowledge learned from multiple source domains to a target domain by dynamically aligning moments of the feature distributions. In this paper, we propose a multi-discriminator active adversarial network to improve the classification performance for multi-center brain disease diagnosis.

III. MATERIALS AND METHOD

In this section, we first introduce the materials used in this work, and then present the proposed approach.

A. Materials

1) *Data Acquisition*: We employ five schizophrenia fMRI datasets in this paper: Nanjing Brain Hospital (NBH) dataset, the Center for Biomedical Research Excellence (COBRE) dataset, the Nottingham dataset, the Taiwan dataset, and the Xiangya dataset. The subjects have the following requirements: (1) No other DSM-IV disease exists; (2) No history of drug abuse; (3) No clinically significant head trauma. The demographic information is summarized in Table I.

2) Data Preprocessing:

a) *NBH and COBRE*: The rs-fMRI images of each participant were acquired by using a 3-Tesla Siemens Tim-Trio scanner with an 8 or 12-channel head coil. During the scanning of images, all participants must keep their eyes open and stare intently at the fixed cross in the middle of the screen for 5 minutes.

b) *Nottingham*: The images were acquired using a 3-T Philips Achieva MRI scanner. During the scanning of images, all participants must keep their eyes open and stare intently at the fixed cross in the middle of the screen for 5 minutes.

c) *Taiwan and Xiangya*: The resting-state fMRI (Rs-fMRI) images of each participant were acquired by using a 3-Tesla Siemens Tim-Trio scanner with an 8 or 12-channel head coil. During the scanning of images, the participants were instructed to keep their eyes closed but not fall asleep.

The first ten volumes for all datasets to be preprocessed were discarded to ensure scanner stabilization and adaptability to the subjects' environment. The remaining functional scans were

TABLE II
DEMOGRAPHIC CHARACTERISTICS OF PARTICIPANTS

Dataset	Patient	Normal	Features
NBH	24	21	90*230
COBRE	53	67	90*145
Nottingham	32	36	90*235
Taiwan	69	62	90*170
Xiangya	83	60	90*170

first corrected to the difference in acquisition time between on-chip scans and then readjusted to the intermediate volume to correct head movement between scans. The functional scan was spatially normalized to the SPM5 Montreal Neurological Institute template and resampled to $3 \times 3 \times 3 \text{ mm}^3$ voxels. After linear detrending, data were filtered using a typical temporal bandpass (0.01–0.08 Hz) to reduce low-frequency drift and high-frequency physiological noise. Next, the motion parameters, the global mean signal, white matter (WM), and cerebrospinal fluid tissue (CSF) as nuisance covariates were used to reduce the effects of head motion and non-neuronal BOLD fluctuations.

After data preprocessing, the fMRI images were first registered to the same template space using the HAMMER deformation registration method and divided into 90 brain areas or regions of interest (ROIs) using the Anatomical Automatic Labeling (AAL) template. Finally, for each ROI, the average fMRI time series over all voxels was taken as the time series for that ROI. The processed information of all datasets is shown in Table II.

As the study of Meszlényi et al. [36] has proven that combination of different functional connectivity metrics can improve classification accuracy, we use three kinds of different measures as the combined inputs. Suppose $x_i(k)$ denotes the k -th value of the i -th time-series from the corresponding brain region. Dynamic Time Warping (DTW) is a time-series distance measure, which can be used for time-series classification [37] and connectivity-based classification [38]. For example, we can obtain the DTW matrix between the time series x_1 and the time series x_2 by

$$DTW(i, j) = \begin{cases} \|x_1(i), x_2(j)\| + d(i, j), & \text{if } i, j > 1 \\ \|x_1(i), x_2(j)\| + DTW(i, j-1), & \text{if } i = 1, j > 1 \\ \|x_1(i), x_2(j)\| + DTW(i-1, j), & \text{if } i > 1, j = 1 \\ \|x_1(i), x_2(j)\|, & \text{if } i = 1, j = 1 \end{cases} \quad (1)$$

$$d(i, j) = \min\{DTW(i, j-1), DTW(i-1, j), DTW(i-1, j-1)\} \quad (2)$$

where $x_1(i)$ denotes the i -th value in time series x_1 and $x_2(j)$ denotes the j -th value in time series x_2 . The minimal value of this matrix is the DTW distance between x_1 and x_2 . In addition, the covariance measures and correlation measures are also commonly used metrics to evaluate functional connectivity

by

$$Cov(x_1, x_2) = E(x_1 x_2) - E(x_1)E(x_2), E(x_1 x_2) = \frac{x_1 x_2^T}{90} \quad (3)$$

$$Cor(x_1, x_2) = \frac{Cov(x_1, x_2)}{\sqrt{V(x_1)}\sqrt{V(x_2)}} \quad (4)$$

where $E(x)$ denotes the mathematical expectation of x and $V(x)$ denotes the variance of x . Finally, we acquire $3 \times 90 \times 90$ features for each sample by using above 3 metrics.

B. Multi-Discriminator Active Adversarial Network

In this paper, we develop a multi-center fusion approach that counters the uncertainties at center, feature, and label levels simultaneously. In the multi-center classification task, we are first given labeled multi-center data from source domain $X^S = \{X^1, X^2, \dots, X^m\}$, where m is the number of domains. Each center data $X^i = \{x_1^i, x_2^i, \dots, x_{n_i}^i\}$ denotes the samples in the i -th center in source domain, where n_i is the number of subjects and $X^i \sim P^i$ ($P^i \neq P^j$, if $i \neq j$). Meanwhile, there is an unlabeled target domain data $X^t = \{x_1^t, x_2^t, \dots, x_{n_t}^t\}$.

The main purpose of this work is to alleviate the problems of data distribution shift among centers, negative transfer impact of noised samples, and lack of supervised information. To this end, we proposed a multi-discriminator active adversarial network (MDAAN) model shown in Fig. 2. It simultaneously extracts the deep latent common feature of multiple centers and fuses the multiple discriminators to obtain invariant representation by adversarial learning. First, we employ the source center feature f^i and the target center feature f^t extracted by a shared feature extractor G to train the discriminator D_i . Then the loss of different centers is combined to train the feature extractor and discriminators by adversarial learning strategy. Finally, we use the information from the discriminators and classifier to label the most valuable samples and retrain the model with a small labeling cost.

1) *Multi-Discriminator Adversarial Domain Adaptation*: One of the cores of learning multi-center invariant representation is to align the domain shift between different centers. It achieves feature alignment across different data domains by adversarial networks and evaluates the contribution of each source domain model to the target domain model.

a) *Deep Latent Feature Extraction*: For multi-center data, we share one feature extractor G to learn the low-dimensional latent representation of each center, given by $G(x)$. Hence, in the case of a single center, the classifier C outputs a SoftMax prediction distribution over the classes, and it is trained using a cross-entropy (CE) loss on the labeled source data given by

$$\mathcal{L}_C^i = -y^i \cdot \log [C(G(x^i))] \quad (5)$$

where y^i is the ground truth label of the sample x^i .

b) *Adversarial Domain Adaptation*: As we described above, the classifier trained only on source data often has an inferior performance for the target data due to the different distribution $P_s \neq P_t$. For this reason, a domain discriminator

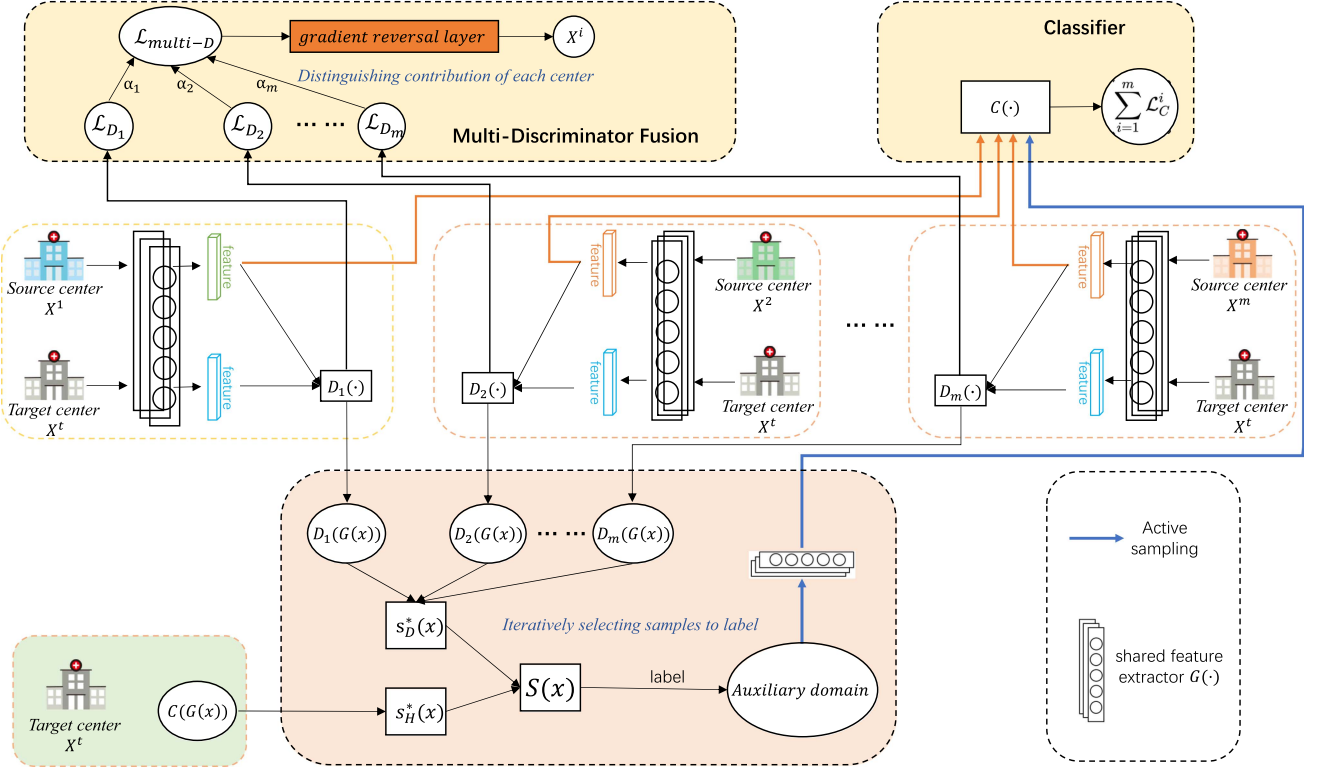


Fig. 2. The proposed multi-discriminator active adversarial network (MDAAN) for multi-center brain disease diagnosis. First, we learn a discriminator for each source center with the target center, and all centers share a feature extractor and a classifier. Then, the multi-discriminator fusion module is used to learn the common subspace, which can adaptively allocate weights for different centers to reduce domain shift in multi-source. Finally, we select the most valuable samples through active learning to alleviate the negative transfer and labeling cost.

D is trained to identify the source and target sample's domain, while G is synchronously trained to generate features to confuse the discriminator by $\mathcal{L}_{D_i}$. The loss is shown as follows

$$\mathcal{L}_{D_i} = -d \cdot \log D_i(G(x)) - (1-d) \cdot \log(1 - D_i(G(x))) \quad (6)$$

where $d = 1$ when x from the target domain, and $d = 0$ when x from source domain.

Then, domain invariant features can be acquired through min-max training. Nevertheless, the adversarial learning strategy only learns the invariant representation between the source domain and the target domain, without considering the inherent correlation information of multi-center data in the source domain. It is well known that directly combining the multiple centers cannot provide an optimal model due to the data uncertainty in different centers. Based on this, we introduce a discriminator for each center of the source domain to alleviate the domain shift, which can learn domain invariant features for each center with the target domain.

c) Center Contribution Allocation: Similar to the phenomenon of negative transfer, not all source domains have equal contribution to the target domain. In some extreme cases, some source domains may be harmful to knowledge transfer. Therefore, it needs to accomplish a validity analysis of each source domain in the process of learning.

We develop the following measure to evaluate the importance of the j -th source center to the target center

$$\alpha_j = \frac{\frac{n_s^j}{n_s} + \frac{\mathcal{L}_{D_j}}{\sum_{j=1}^N \mathcal{L}_{D_j}}}{2} \quad (7)$$

where n_s is the number of the subjects of all the source centers, n_s^j is the number of the subjects of j -th source center, and $\mathcal{L}_{D_j}$ is the discriminator loss of j -th center.

The above indicator for measuring the importance of the model on each domain mainly considers the following two aspects. On one hand, when the shared feature extractor is fixed, the higher discriminator loss $\mathcal{L}_D$ means better effectiveness in learning the common subspace between the source center and the target center. On the other hand, Ben-David et al. [39] found that the optimal α should be proportionate to the amount of data in transfer learning. Hence, we use both the discriminator loss $\mathcal{L}_{D_j}$ and the amount of data n_s^j to evaluate the domain weight.

Meanwhile, we use a classifier C trained on all centers from source domain and target domain to improve the discriminability of the model during the adversarial training. The loss of discriminators on all centers is given by:

$$\mathcal{L}_{multi-D} = \frac{1}{m} \sum_{i=1}^m \alpha_i \cdot \mathcal{L}_{D_i} \quad (8)$$

The overall loss of the proposed network model is as follows:

$$\mathcal{L}_{MDA} = \frac{1}{m} \sum_{i=1}^m \mathcal{L}_C^i - \beta \cdot \mathcal{L}_{multi-D} \quad (9)$$

where β is the trade-off parameter to control the contribution of $\mathcal{L}_C$ and $\mathcal{L}_{multi-D}$.

2) *Active Labeling in Target Domain*: In order to improve classification performance with limited labeling cost, we adaptively select the most valuable subjects to label by introducing active learning in above model.

We aim to label the most hard classified and hard aligned samples and add them in model training, which can promote the alignment between the source centers and the target center. In active learning, the value of sample labeling can be measured by error reduction [40], [41], query by committee [42], uncertainty [43], etc. As entropy is a natural measure of uncertainty, in our work, we use it to evaluate the classification difficulty of the sample x by

$$s_H^*(x) = - \sum_i P_\theta(y_i|x) \log P_\theta(y_i|x) \quad (10)$$

where y_i ranges over all possible labels, θ denotes the parameters of classifier C . The larger the value of $s_H^*(x)$ is, the more difficult it is to determine the category of x by classifier C .

In addition, we also need to select the hard fused subjects to label, which could help avoid the negative transferred impact of hard learning samples to a certain extent. For example, Shi et al. [44] regarded the prediction of the discriminator as the quantification for the domain similarity of each sample. In adversarial domain adaptation, the samples far from the decision boundary of discriminator can be recognized as the hard confused samples, while samples close to the decision boundary are easy confused samples [45], [46]. Specifically, we use the output of multiple discriminators for each sample to choose those samples that are still hard to distinguish by well-trained discriminators, which is given by

$$s_D^*(x) = \sum_{i=1}^m |D_i(G(x)) - 0.5| \quad (11)$$

where D_i denotes the i -th discriminator. $s_D^*(x)$ is used to denote the transfer difficulty of the sample x . If the value of x is high, it means that the features of the sample x are difficult to align with the target center.

Finally, we use the following score to determine which samples need to label.

$$s(x) = s_D^*(x) + \gamma s_H^*(x) \quad (12)$$

We use the trade-off parameter γ to balance these two terms in (12).

After calculating the above score for each sample in the target domain, we select the top k samples with the highest $s(x)$ values to be included in the auxiliary domain, participating in subsequent model training.

3) *Uncertainty Mitigation in the Proposed Multi-Center Analysis*: Our model uses domain adversarial training network architecture to alleviate the uncertainties at center, feature, and

Algorithm 1: The Proposed MDAAN Algorithm.

Input: Labeled multi-center source dataset

$X^S = \{X^1, X^2, \dots, X^m\}$

Labeled target auxiliary dataset $X^{lt} = \emptyset$

Unlabeled target dataset X^t

Output: Feature extractor G , Classifier C and

Discriminators $\{D_i\}_{i=1}^m$

1: Initialize $G, C, \{D_i\}_{i=1}^m$

2: **for** Shotnum to MaxShotnum **do**

3: $f^i = G(X^i), f^t = G(X^t)$ and $f^{lt} = G(X^{lt})$

4: $\mathcal{L}_C^i \leftarrow C(f^i, f^{lt})$, calculated by (5)

5: $\mathcal{L}_{D_i} \leftarrow D_i(f^i, f^{lt}; f^t)$, calculated by (6)

6: $\alpha_i \leftarrow \mathcal{L}_{D_i}$, calculated by (7)

7: $\mathcal{L}_{multi-D} \leftarrow \alpha_i, \mathcal{L}_{D_i}$, calculated by (8)

8: $\mathcal{L}_{MDA} \leftarrow \mathcal{L}_C^i, \mathcal{L}_{multi-D}$, calculated by (9)

9: Select the most hard classified and hard aligned samples to label by $s(x)$ in (12), and add these samples to X^{lt}

10: **end for**

label levels. At feature level, we use a share feature extractor to extract domain invariant features among centers, which transfers target center samples to a distribution close to the source centers as much as possible. At center level, we adaptively estimate the contribution of each center in fusion to reduce the influence of centers with large data distribution differences on modeling. At label level, we select the most valuable samples to label and use them as the auxiliary domain to improve the robustness of the model and greatly reduce the labeling cost. The proposed approach is a novel solution to multi-level uncertainties for the multi-center brain disease diagnosis. The algorithm is briefly summarized in Algorithm 1.

IV. EXPERIMENTS

In this section, we first introduce the competing methods and the experimental setup. Then we present our experimental results on the real multi-center schizophrenia dataset, as well as the comparison between our method and several state-of-the-art methods. The code of the proposed MDAAN is available at: <https://github.com/whyiswmm/MDAAN>.

A. Competing Methods

In this experiment, we compare the proposed method with the following seven single-source domain adaptation methods, including Domain-Adversarial Neural Networks (DANN) [47], Conditional adversarial domain adaptation (CDAN) [27], Deep Adaptation Networks (DAN) [24], Transfer Component Analysis (TCA) [48], Instance Level Affinity-Based Transfer for Unsupervised Domain Adaptation (ILA-DA) [49], Safe Self-Refinement for Transformer-Based Domain Adaptation (SSRT) [50], Minimum Class Confusion (MCC) [51], and three multi-source domain adaptation methods, including Multi-source Distilling Domain Adaptation (MDDA) [28], Moment Matching for Multi-Source Domain Adaptation (M3SDA) [35],

Pseudo Target for MDA (PTMDA) [52]. For these deep learning methods, the connectome-convolutional neural network (CCNN) [36] has been used for extracting the feature.

a) *DANN*: DANN is one of the most widely used adversarial domain adaptation framework. It trains a discriminator for adversarial training and a classifier for making decision. Specifically, it combines all source centers as the whole source domain and learns a discriminative model with both source and target domain to obtain the final classification result.

b) *CDAN*: CDAN is an adversarial adaptation model with discriminative information conveyed in the classifier predictions. There are two conditioning strategies in CDAN: multi-linear conditioning that captures the cross-covariance between feature representations and classifier predictions, and entropy conditioning that controls the uncertainty of classifier predictions transferability.

c) *DAN*: In this method, the hidden representations of all task-specific layers are embedded in a reproducing kernel Hilbert space, where the mean embeddings of different domain distributions can be explicitly matched. Furthermore, the domain discrepancy is further reduced using an optimized multi-kernel selection scheme for mean embedding matching.

d) *TCA*: The TCA method tries to learn several transfer components across domains in a Reproducing Kernel Hilbert Space (RKHS) using Maximum Mean Discrepancy, to make data distributions in different domains to be close to each other. In TCA, we learn the common feature representation in source and target domains, and then the KNN is used as the classifier to provide the final prediction of the target data.

e) *ILA-DA*: ILA-DA is a method that simultaneously considers intra-class clustering as well as inter-class separation across datasets to improve unsupervised domain adaptation. It proved to be a credible and efficient method to extract similar and dissimilar samples across source domain and target domain and utilize a multi-sample contrastive loss to drive the domain alignment process.

f) *SSRT*: It takes a vision transformer as the backbone network and utilizes predictions on perturbed target domain data to refine the adapted model. SSRT imposes a regularization on the corresponding transformer layers by adding random offsets to the latent token sequences of target domain data and minimizing the discrepancy of model's predicted probabilities between the original and perturbed versions.

g) *MCC*: MCC is a general loss function, which can be characterized as a non-adversarial DA method enjoying faster convergence speed without explicitly deploying domain alignment.

h) *MDDA*: MDDA extracts the different distances among multiple source domains and the target domain, and investigates the different similarities of the source samples to the target ones.

i) *M3SDA*: M3SDA transfers knowledge from multiple labeled source domains to an unlabeled target domain by dynamically aligning their feature distributions through moment matching.

j) *PTMDA*: PTMDA constructs a series of pseudo target domains by performing the adversarial training strategy on the real target domain and each source domain, and introduces

the matching-normalization layer to align the distributions of different domains to improve the transfer ability.

B. Comparison on Multi-Center Diagnosis Task

To evaluate the classification performance, we use 7 evaluation metrics, including classification accuracy (ACC), sensitivity (SEN), specificity (SPE), balanced accuracy (BAC), positive predictive value (PPV), negative predictive value (NPV), and the area under the receiver operating characteristic (ROC) curve (AUC) [53], which have been widely used in diagnosis performance analysis. In detail, TP, TN, FP and FN denote True Positive, True Negative, False Positive, and False Negative, respectively. The evaluation metrics can be calculated as follows: $ACC = (TP + TN) / (TP + TN + FP + FN)$, $SEN = TP / (TP + FN)$, $SPE = TN / (TN + FP)$, $BAC = (SEN + SPE) / 2$, $PPV = TP / (TP + FP)$, and $NPV = TN / (TN + FN)$. For these metrics, a higher value means better performance. As can be seen from Table III, our method performs better than the comparison method in most cases. Specifically, our proposed method achieved the highest performance in metrics of ACC, AUC, and BAC among all the methods for all target domains. For instance, in the NBH target domain, although the multi-source learning approach of M3SDA achieves slightly better SPE than our method, its SEN value is significantly lower than our method by more than 12%, and its other metrics are also inferior to our method.

Next, we compare the performance of all methods on cross-center setting, and evaluate the individual diagnostic classification of our model on the schizophrenia dataset. Specifically, we choose one center as the target domain, and one of the remaining centers is treated as the source domain, which can be viewed as single source domain learning problem. It is worth noting that the multi-source learning approaches, such as MDDA, M3SDA, PTMDA, cannot be applied to single source domain learning task. The results have been shown in Table IV. The results show that our method still achieves the best performance in most cases, which suggests that the treatment of our method in multi-center data uncertainty is helpful to improve the diagnostic accuracy.

C. Ablation Experiment

In this subsection, we will measure the contribution of the main modules of our proposed method, including the multi-discriminator adversarial domain adaptation module, center weighting module, and the sample labeling value estimation module. Removing the multi-discriminator adversarial domain adaptation module from our algorithm means that the model is only based on the mixed domains without considering the feature representation gap among domains. Removing the center weighting module means that the weight of each center is equal in the multi-discriminator adversarial domain adaptation module. When the active learning module is absent during the training process, our method will degenerate into a conventional unsupervised algorithm and it cannot make full use of labeling resources. We use the experiment setting in Section IV-B, and the ablation experiment results are shown in Table V. Obviously, the results suggest each module is helpful for diagnosis performance promotion. More importantly, our algorithm achieves higher diagnostic accuracy than its degraded versions.

TABLE III
THE CLASSIFICATION RESULTS (%) OF DIFFERENT DOMAIN ADAPTATION MODELS ON THE MULTI-CENTER SCHIZOPHRENIA DIAGNOSIS TASK

Target domain	Method	ACC	SEN	SPE	AUC	BAC	PPV	NPV
NBH	DANN	82.22	91.67	71.43	80.16	81.55	78.57	88.24
	CDAN	72.73	70.83	75.00	86.56	72.92	77.27	68.18
	DAN	57.78	62.50	52.38	62.40	57.44	60.00	55.00
	TCA	64.44	75.00	52.38	63.69	63.69	64.29	64.71
	ILA-DA	82.22	87.50	76.19	81.45	81.85	80.77	84.21
	SSRT	66.67	70.83	61.90	74.60	66.37	68.00	65.00
	MCC	73.33	83.33	61.90	73.61	72.62	71.43	76.47
	MDDA	82.93	90.00	76.19	90.95	83.10	78.26	88.89
	M3SDA	82.22	83.33	80.95	87.10	82.14	83.33	80.95
	PTMDA	84.21	94.12	76.19	92.44	85.15	76.19	93.33
	Ours	88.37	95.83	80.00	92.71	87.92	85.19	94.12
COBRE	DANN	64.17	96.23	38.81	58.55	67.52	55.43	92.86
	CDAN	64.17	56.60	70.15	77.77	63.38	60.00	67.14
	DAN	54.17	49.06	58.21	54.17	53.63	48.15	59.09
	TCA	61.67	67.92	56.72	62.32	62.32	55.38	69.09
	ILA-DA	62.50	49.06	73.13	64.05	61.10	59.09	64.47
	SSRT	68.33	54.72	79.10	68.80	66.91	67.44	68.83
	MCC	66.67	47.17	82.09	60.24	64.63	67.57	66.27
	MDDA	72.97	66.67	77.78	78.87	72.22	69.57	75.38
	M3SDA	73.33	56.60	86.57	78.60	71.59	76.92	71.60
	PTMDA	74.11	61.22	84.13	80.05	72.68	75.00	73.61
	Ours	78.99	88.68	71.21	82.10	79.95	71.21	88.68
Nottingham	DANN	75.00	84.38	66.67	77.34	75.52	69.23	82.76
	CDAN	70.59	56.25	83.33	85.33	69.79	75.00	68.18
	DAN	50.00	43.75	55.56	52.91	49.65	46.67	52.63
	TCA	58.82	68.75	50.00	59.38	59.38	55.00	64.29
	ILA-DA	76.47	68.75	83.33	77.43	76.04	78.57	75.00
	SSRT	64.71	65.62	63.89	65.89	64.76	61.76	67.65
	MCC	73.53	81.25	66.67	74.65	73.96	68.42	80.00
	MDDA	75.00	68.75	80.56	79.77	74.65	75.86	74.36
	M3SDA	76.47	71.88	80.56	76.30	76.22	76.67	76.32
	PTMDA	77.42	65.52	87.88	79.10	76.70	82.61	74.36
	Ours	86.57	87.50	85.71	87.86	86.61	84.85	88.24
Taiwan	DANN	71.76	65.22	79.03	75.29	72.12	77.59	67.12
	CDAN	70.77	53.62	90.16	85.21	71.89	86.05	63.22
	DAN	54.20	47.83	61.29	49.03	54.56	57.89	51.35
	TCA	67.94	66.67	69.35	68.01	68.01	70.77	65.15
	ILA-DA	67.94	71.01	64.52	70.76	67.77	69.01	66.67
	SSRT	74.81	72.46	77.42	78.21	74.94	78.12	71.64
	MCC	61.07	65.22	56.45	56.29	60.83	62.50	59.32
	MDDA	79.39	89.86	67.74	85.74	78.80	75.61	85.71
	M3SDA	79.51	76.47	83.33	85.05	79.90	85.25	73.77
	PTMDA	80.47	78.26	83.05	85.48	80.66	84.38	76.56
	Ours	83.46	92.75	72.88	85.75	82.82	80.00	89.58
Xiangya	DANN	61.54	53.01	73.33	59.08	63.17	73.33	53.01
	CDAN	54.93	60.24	47.46	60.35	53.85	61.73	45.90
	DAN	55.94	51.81	61.67	50.88	56.74	65.15	48.05
	TCA	53.85	51.81	56.67	54.24	54.24	62.32	45.95
	ILA-DA	60.84	54.22	70.00	59.44	62.11	71.43	52.50
	SSRT	62.94	65.06	60.00	61.81	62.53	69.23	55.38
	MCC	60.84	79.52	35.00	53.92	57.26	62.86	55.26
	MDDA	63.64	55.42	75.00	63.29	65.21	75.41	54.88
	M3SDA	64.79	66.27	62.71	64.35	64.49	71.43	56.92
	PTMDA	64.54	74.39	50.85	63.87	62.62	67.78	58.82
	Ours	70.59	74.07	65.45	72.64	69.76	75.95	63.16

D. Parameter Analysis

We also analyze the effect of the hyper-parameters β and γ for the performance of our proposed method. Here, each center is alternatively selected as the target center, and the remaining centers are treated as source centers. Specifically, we independently change the values of β and γ . β is selected from $\{0.1, 0.2, 0.3, 0.4, 0.5\}$, while γ is chosen from $\{0.2, 0.4, 0.6, 0.8, 1.0\}$. Fig. 3 demonstrates that the classification results of MDAAN are stable for γ and β in most cases, indicating that our method is robust to these two parameters.

E. Comparison With Different Active Learning Strategies

In this experiment, we compare our method with several widely used active learning strategies in choosing the most

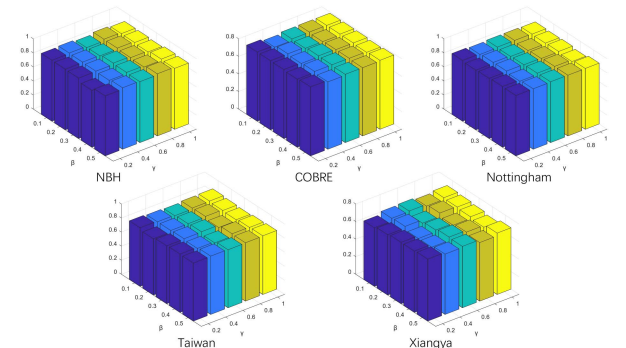


Fig. 3. The classification accuracy of our proposed method with different combinations of the hyper-parameters β and γ .

TABLE IV
THE CLASSIFICATION ACCURACIES (%) OF DIFFERENT DOMAIN ADAPTATION METHODS ON THE CROSS-CENTER SCHIZOPHRENIA DIAGNOSIS TASK

Target Domain	Method	NBH	COBRE	Nottingham	Taiwan	Xiangya	Avg
NBH	DANN	-	71.11	71.11	73.33	71.11	71.67
	FADA	-	61.36	61.64	63.64	55.88	60.63
	DAN	-	44.44	48.89	60.00	57.78	52.78
	TCA	-	44.44	68.89	68.89	55.56	59.45
	ILA-DA	-	66.67	53.33	53.33	53.33	56.67
	SSRT	-	62.22	66.67	60.00	66.67	63.89
	MCC	-	68.89	66.67	71.11	62.22	67.22
	Ours	-	81.40	79.07	81.40	83.33	81.28
COBRE	DANN	59.17	-	67.50	64.17	54.17	61.25
	CDAN	52.50	-	58.33	64.17	55.00	57.5
	DAN	50.00	-	55.00	51.67	53.33	52.5
	TCA	57.50	-	55.00	67.50	54.17	58.54
	ILA-DA	44.17	-	59.17	55.83	55.83	53.75
	SSRT	56.67	-	67.50	60.83	55.83	60.21
	MCC	58.33	-	60.83	59.17	59.17	59.38
	Ours	71.19	-	75.42	75.42	55.08	69.28
Nottingham	DANN	58.82	67.65	-	66.18	60.29	63.24
	CDAN	52.94	69.12	-	69.12	66.18	64.34
	DAN	50.00	51.47	-	61.76	58.82	55.51
	TCA	60.29	57.35	-	61.76	50.00	57.35
	ILA-DA	51.47	55.88	-	55.88	64.71	56.99
	SSRT	60.29	61.76	-	66.18	58.82	61.76
	MCC	63.24	63.24	-	67.65	66.18	65.08
	Ours	75.76	77.27	-	75.38	71.21	74.91
Taiwan	DANN	70.99	70.99	65.65	-	61.83	67.37
	CDAN	63.85	62.31	60.77	-	68.46	63.85
	DAN	54.20	57.25	54.20	-	48.09	53.44
	TCA	70.23	66.41	70.99	-	48.85	64.12
	ILA-DA	52.67	53.44	52.67	-	53.44	53.06
	SSRT	67.18	71.76	61.83	-	61.07	65.46
	MCC	61.83	59.54	58.78	-	63.36	60.88
	Ours	79.07	79.07	79.07	-	61.24	74.61
Xiangya	DANN	60.84	52.45	55.94	62.94	-	58.04
	CDAN	58.45	54.93	52.11	57.04	-	55.63
	DAN	46.85	55.94	52.45	58.74	-	53.50
	TCA	59.44	39.86	54.55	57.34	-	52.80
	ILA-DA	58.04	58.04	58.74	58.74	-	58.39
	SSRT	58.04	58.04	58.74	59.44	-	58.57
	MCC	58.74	51.75	56.64	60.84	-	56.99
	Ours	66.67	65.25	61.70	63.83	-	64.39

TABLE V
ABLATION EXPERIMENT OF THE PROPOSED METHOD

Multi-discriminator adversarial domain adaptation	Center weighting	Sample labelling value estimation	Target domain				
			NBH	COBRE	Nottingham	Taiwan	Xiangya
×	×	×	73.33	69.17	66.18	67.94	59.44
×	×	✓	80.00	73.28	71.88	70.23	61.59
×	✓	×	75.56	71.67	67.65	69.47	60.84
×	✓	✓	82.22	73.33	73.53	71.76	62.94
✓	×	×	77.78	71.67	70.59	69.47	60.84
✓	×	✓	84.44	74.17	72.06	73.28	65.03
✓	✓	×	82.22	75.83	80.88	75.57	61.54
✓	✓	✓	88.37	78.99	86.57	83.46	70.59

valuable samples to label, including Query by Committee (QBC) [42], Querying Informative and Representative Examples (QUIRE) and Random QUIRE. In addition, we also compare our method with its unsupervised version that trains the model without an active learning strategy.

Fig. 4 shows the comparison of our method and several active learning methods in multi-center scenarios. As can be seen in Fig. 4, the random selection method is unstable because it cannot guarantee that the selected samples are favorable for model

transfer. Most results show that the effect of active learning is better than random selection. It means active learning is beneficial to the multi-center task. Moreover, the active learning strategy we used is more effective than the comparison methods, because it identifies the hard classified and hard aligned samples to label to improve the classification performance in the multi-center diagnosis problem.

We further validate the proposed active learning strategy on different domain adaption methods. In this experiment, we

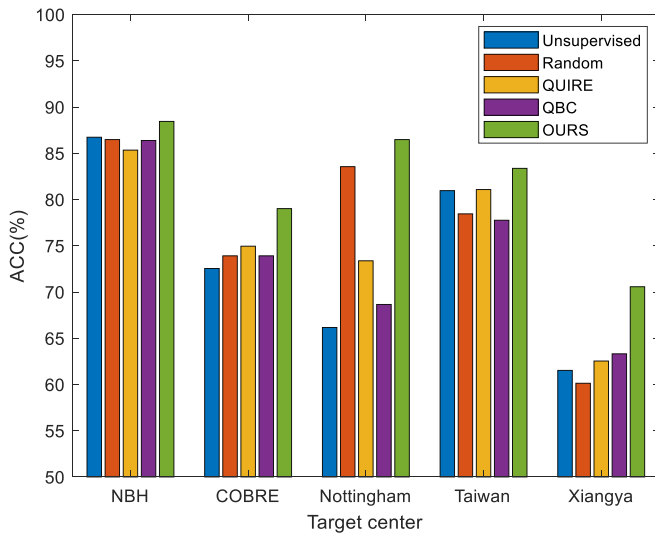


Fig. 4. The comparison of our active learning strategy and several comparison methods in multi-center settings.

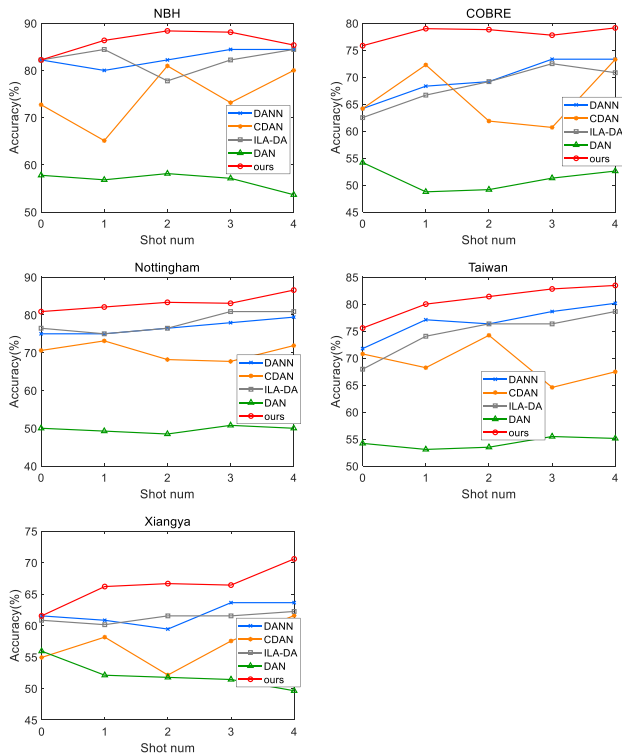


Fig. 5. The results of the proposed active learning strategy and comparison methods on different multi-center tasks.

consider comparing the proposed approach with a deep domain adaptation method DAN and three adversarial domain adaptation methods including DANN, CDAN, and ILA-DA. The experimental results are shown in Fig. 5, in which the shot num means the number of selected samples in each training process. In this experiment, we report the accuracies of these methods with different number of labeling subjects in the center. In general, our method performs better than other domain adaption

methods with the proposed active learning strategy. When shot num being zero, the diagnosis task becomes conventional UDA problem. In most centers, the accuracies are improved with the increase of shot num. However, the performance of the model has decreased in some cases (such as the shot num 4, site NBH). We believe it is due to some special samples that lead to the contamination of the training dataset. Compared with our proposed method, other domain adaptation methods contain less discriminative information in sampling labeling value estimation module, which leads to an unstable results. Experimental results also show that our proposed method is superior to the state-of-the-art UDA methods with low labeling cost.

V. CONCLUSION

The multi-center approach has been widely used in brain disease diagnosis, and the uncertainty in multi-center is more common than that in single center. This paper aims at alleviating the influence of uncertainties of multi-center fusion in different aspects and improving the diagnostic performance under small annotation cost. To this end, we propose a multi-discriminator active adversarial network, and apply it into multi-site brain disease diagnosis. In the proposed method, an adversarial learning framework based on a generative adversarial network with multi-discriminator is developed to handle the domain shift among different centers. Meanwhile, it adaptively fuses these discriminators from source centers according to their distribution differences with target center. Then, the hard classified and hard aligned samples are identified to label by active learning. Finally, we treated the selected samples as the auxiliary domain to improve the robustness of the model. The experimental results proved that the proposed method outperforms several state-of-the-art multi-center analysis methods.

REFERENCES

- [1] Y. I. Sheline, "Neuroimaging studies of mood disorder effects on the brain," *Biol. Psychiatry*, vol. 54, no. 3, pp. 338–352, 2003.
- [2] P. Padilla, M. López, J. M. Górriz, J. Ramirez, D. Salas-Gonzalez, and I. Alvarez, "NMF-SVM based CAD tool applied to functional brain images for the diagnosis of Alzheimer's disease," *IEEE Trans. Med. Imag.*, vol. 31, no. 2, pp. 207–216, Feb. 2012.
- [3] X. Zhu, H.-I. Suk, H. Huang, and D. Shen, "Low-rank graph-regularized structured sparse regression for identifying genetic biomarkers," *IEEE Trans. Big Data*, vol. 3, no. 4, pp. 405–414, Dec. 2017.
- [4] A. Phinyomark, E. Ibáñez-Marcelo, and G. Petri, "Resting-state fMRI functional connectivity: Big data preprocessing pipelines and topological data analysis," *IEEE Trans. Big Data*, vol. 3, no. 4, pp. 415–428, Dec. 2017.
- [5] M. Wang, J. Huang, M. Liu, and D. Zhang, "Modeling dynamic characteristics of brain functional connectivity networks using resting-state functional mri," *Med. Image Anal.*, vol. 71, 2021, Art. no. 102063.
- [6] J. Huang, M. Wang, X. Xu, B. Jie, and D. Zhang, "A novel node-level structure embedding and alignment representation of structural networks for brain disease analysis," *Med. Image Anal.*, vol. 65, 2020, Art. no. 101755.
- [7] P. Dluhoš et al., "Multi-center machine learning in imaging psychiatry: A meta-model approach," *Neuroimage*, vol. 155, pp. 10–24, 2017.
- [8] Y. Xu et al., "Data uncertainty in face recognition," *IEEE Trans. Cybern.*, vol. 44, no. 10, pp. 1950–1961, Oct. 2014.
- [9] Y. Song, G. Zhang, J. He, S. Liao, and N. Xie, "Uncertainty measurement for heterogeneous data: An application in attribute reduction," *Artif. Intell. Rev.*, vol. 55, no. 2, pp. 991–1027, 2022.
- [10] M. Leming, J. M. Górriz, and J. Suckling, "Ensemble deep learning on large, mixed-site fMRI datasets in autism and other tasks," *Int. J. Neural Syst.*, vol. 30, no. 07, 2020, Art. no. 2050012.

- [11] A. S. Heinsfeld, A. R. Franco, R. C. Craddock, A. Buchweitz, and F. Meneguzzi, "Identification of autism spectrum disorder using deep learning and the abide dataset," *Neuroimage Clin.*, vol. 17, pp. 16–23, 2018.
- [12] M. Wang, D. Zhang, J. Huang, P.-T. Yap, D. Shen, and M. Liu, "Identifying autism spectrum disorder with multi-site fmri via low-rank domain adaptation," *IEEE Trans. Med. Imag.*, vol. 39, no. 3, pp. 644–655, Mar. 2020.
- [13] C. Shi, X. Xin, and J. Zhang, "Domain adaptation using a three-way decision improves the identification of autism patients from multisite fmri data," *Brain Sci.*, vol. 11, no. 5, 2021, Art. no. 603.
- [14] G. Wilson and D. J. Cook, "A survey of unsupervised deep domain adaptation," *ACM Trans. Intell. Syst. Technol.*, vol. 11, no. 5, pp. 1–46, 2020.
- [15] Y. Zhang, Y. Zhang, Y. Wang, and Q. Tian, "Domain-invariant adversarial learning for unsupervised domain adaption," 2018, *arXiv: 1811.12751*.
- [16] M. H. Lee, C. D. Smyser, and J. S. Shimony, "Resting-state fMRI: A review of methods and clinical applications," *Amer. J. Neuroradiol.*, vol. 34, no. 10, pp. 1866–1872, 2013.
- [17] Q. Zhu, J. Huang, and X. Xu, "Non-negative discriminative brain functional connectivity for identifying schizophrenia on resting-state fMRI," *Biomed. Eng. Online*, vol. 17, no. 1, pp. 1–15, 2018.
- [18] S. H. Hojjati, A. Ebrahimzadeh, and A. Babajani-Feremi, "Identification of the early stage of alzheimer's disease using structural MRI and resting-state fMRI," *Front. Neurol.*, vol. 10, 2019, Art. no. 904.
- [19] Y. Li et al., "Multimodal hyper-connectivity of functional networks using functionally-weighted lasso for mci classification," *Med. Image Anal.*, vol. 52, pp. 80–96, 2019.
- [20] R. Ju et al., "Early diagnosis of alzheimer's disease based on resting-state brain networks and deep learning," *IEEE/ACM Trans. Comput. Biol. Bioinf.*, vol. 16, no. 1, pp. 244–257, Jan./Feb. 2019.
- [21] Q. Zhu, J. Yang, S. Wang, D. Zhang, and Z. Zhang, "Multi-modal non-euclidean brain network analysis with community detection and convolutional autoencoder," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 7, no. 2, pp. 436–446, Apr. 2023.
- [22] W. M. Kouw and M. Loog, "A review of domain adaptation without target labels," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 3, pp. 766–785, Mar. 2021.
- [23] M. Long, J. Wang, J. Sun, and S. Y. Philip, "Domain invariant transfer kernel learning," *IEEE Trans. Knowl. Data Eng.*, vol. 27, no. 6, pp. 1519–1532, Jun. 2015.
- [24] M. Long, Y. Cao, J. Wang, and M. Jordan, "Learning transferable features with deep adaptation networks," in *proc. Int. Conf. Mach. Learn.*, PMLR, 2015, pp. 97–105.
- [25] I. Goodfellow et al., "Generative adversarial nets," in *Proc. Adv. Neural Inf. Process. Syst.*, 2014, pp. 2672–2680.
- [26] E. Tzeng, J. Hoffman, K. Saenko, and T. Darrell, "Adversarial discriminative domain adaptation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 7167–7176.
- [27] M. Long, Z. Cao, J. Wang, and M. I. Jordan, "Conditional adversarial domain adaptation," 2017, *arXiv: 1705.10667*.
- [28] S. Zhao et al., "Multi-source distilling domain adaptation," in *Proc. AAAI Conf. Artif. Intell.*, 2020, pp. 12 975–12 983.
- [29] X. Li, Y. Xu, N. Li, B. Yang, and Y. Lei, "Remaining useful life prediction with partial sensor malfunctions using deep adversarial networks," *IEEE/CAA J. Automatica Sinica*, vol. 10, no. 1, pp. 121–134, Jan. 2023.
- [30] H. Tang and K. Jia, "Discriminative adversarial domain adaptation," in *Proc. AAAI Conf. Artif. Intell.*, 2020, pp. 5940–5947.
- [31] N. Xiao and L. Zhang, "Dynamic weighted learning for unsupervised domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 15 242–15 251.
- [32] Z. Gao, S. Zhang, K. Huang, Q. Wang, and C. Zhong, "Gradient distribution alignment certificates better adversarial domain adaptation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 8937–8946.
- [33] S.-J. Huang and S. Chen, "Transfer learning with active queries from source domain," in *Proc. Int. Joint Conf. Artif. Intell.*, 2016, pp. 1592–1598.
- [34] W. Zhang, Z. Wang, and X. Li, "Blockchain-based decentralized federated transfer learning methodology for collaborative machinery fault diagnosis," *Rel. Eng. Syst. Saf.*, vol. 229, 2023, Art. no. 108885.
- [35] X. Peng, Q. Bai, X. Xia, Z. Huang, K. Saenko, and B. Wang, "Moment matching for multi-source domain adaptation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2019, pp. 1406–1415.
- [36] R. J. Meszlényi, K. Buza, and Z. Vidnyánszky, "Resting state fMRI functional connectivity-based classification using a convolutional neural network architecture," *Front. Neuroinform.*, vol. 11, 2017, Art. no. 61.
- [37] H. Ding, G. Trajcevski, P. Scheuermann, X. Wang, and E. Keogh, "Querying and mining of time series data: Experimental comparison of representations and distance measures," in *Proc. VLDB Endowment*, vol. 1, no. 2, pp. 1542–1552, 2008.
- [38] R. J. Meszlényi, P. Hermann, K. Buza, V. Gál, and Z. Vidnyánszky, "Resting state fMRI functional connectivity analysis using dynamic time warping," *Front. Neurosci.*, vol. 11, 2017, Art. no. 75.
- [39] S. Ben-David, J. Blitzer, K. Crammer, A. Kulesza, F. Pereira, and V. Jennifer Wortman, "A theory of learning from different domains," *Mach. Learn.*, vol. 79, pp. 151–175, 2010.
- [40] Y. Tang, Y.-B. Yang, and Y. Gao, "Self-paced dictionary learning for image classification," in *Proc. 20th ACM Int. Conf. Multimedia*, 2012, pp. 833–836.
- [41] J. Tang, Z.-J. Zha, D. Tao, and T.-S. Chua, "Semantic-gap-oriented active learning for multilabel image annotation," *IEEE Trans. Image Process.*, vol. 21, no. 4, pp. 2354–2360, Apr. 2012.
- [42] H. S. Seung, M. Oppor, and H. Sompolinsky, "Query by committee," in *Proc. 5th Annu. Workshop Comput. Learn. Theory*, 1992, pp. 287–294.
- [43] Y. Yan and S.-J. Huang, "Cost-effective active learning for hierarchical multi-label classification," in *Proc. Int. Joint Conf. Artif. Intell.*, 2018, pp. 2962–2968.
- [44] W. Shi, L. Zhang, W. Chen, and S. Pu, "Universal Domain Adaptive Object Detector," in *Proc. 30th ACM Int. Conf. Multimedia*, 2022, pp. 2258–2266.
- [45] Z. He, L. Zhang, X. Gao, and D. Zhang, "Multi-adversarial Faster-RCNN with Paradigm Teacher for Unrestricted Object Detection," *Int. J. Comput. Vis.*, vol. 131, no. 3, pp. 680–700, 2022.
- [46] Z. He and L. Zhang, "Multi-adversarial faster-RCNN for unrestricted object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2019, pp. 6668–6677.
- [47] Y. Ganin et al., "Domain-adversarial training of neural networks," *J. Mach. Learn. Res.*, vol. 17, no. 1, pp. 2096–2030, 2016.
- [48] S. J. Pan, I. W. Tsang, J. T. Kwok, and Q. Yang, "Domain adaptation via transfer component analysis," *IEEE Trans. Neural Netw.*, vol. 22, no. 2, pp. 199–210, Feb. 2011.
- [49] A. Sharma, T. Kalluri, and M. Chandraker, "Instance level affinity-based transfer for unsupervised domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 5361–5371.
- [50] T. Sun, C. Lu, T. Zhang, and H. Ling, "Safe self-refinement for transformer-based domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 7191–7200.
- [51] Y. Jin, X. Wang, M. Long, and J. Wang, "Minimum class confusion for versatile domain adaptation," in *Proc. Eur. Conf. Comput. Vis.*, Springer, 2020, pp. 464–480.
- [52] C. Ren, Y. Liu, X. Zhang, and K. Huang, "Multi-source unsupervised domain adaptation via pseudo target domain," *IEEE Trans. Image Process.*, vol. 31, pp. 2122–2135, 2022.
- [53] G. S. Fletcher, *Clinical Epidemiology: The Essentials*. Philadelphia, PA, USA: Lippincott Williams & Wilkins, 2019.